

Ch 5 function & relation

⑦

4
3
2
1
Lobby

$$A \times B = \{ (x, y) \mid \begin{matrix} x \in A \\ y \in B \end{matrix} \}$$

$$|A \times B| = |A| \cdot |B|$$

$$A = \{ \text{甲, 乙, 丙} \}$$

$$B = \{ 1, 2, 3 \}$$

relation: Any subset of $A \times B$

binary relation $A \times A$

R from $A \times B$

$$R \text{ on } A = \{ (x, y) \mid x \leq y \}$$

$$(2, 3) \in R \equiv 2 R 3$$

function

nonempty A, B .

function (mapping) from A to B

$$f: A \rightarrow B$$

① relation

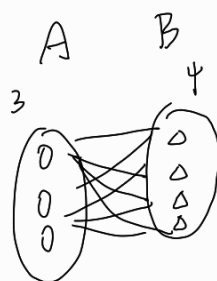
② every element of A

appear exactly once

都要出现

甲只能出现一次

$$\text{ex: } f(a) = b \quad \text{or} \quad (a, b) \in f$$



$$|A \times B| = 12$$

$$f = \{ (\text{甲}, 1), (\text{乙}, 2), (\text{丙}, 1) \}$$

$$f: A \rightarrow B$$

domain codomain

$$f(x) = y \quad (x, y) \in f$$

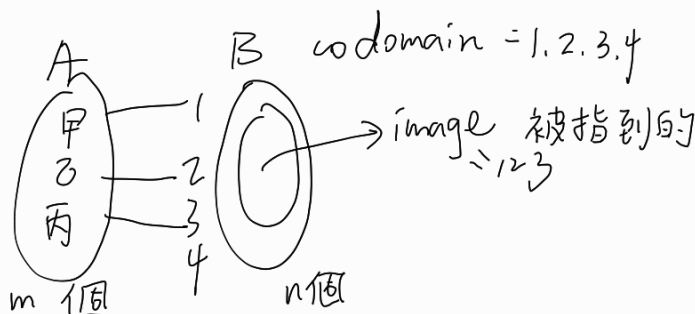


image = codomain $\Rightarrow$ onto function.

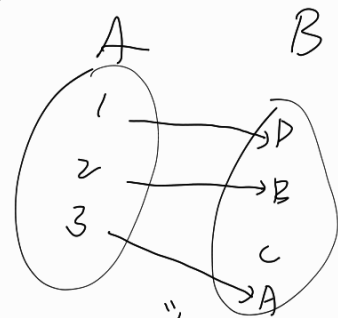
$$|A| = m \quad |A \times B| = m \times n$$

$$|B| = n \quad P(|A \times B|) = 2^{m \times n}$$

選擇 個數 $4 \times 3 \times 2$

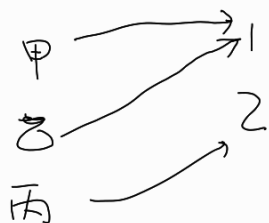
1 to 1 $\longleftrightarrow a_1, a_2 \in A$

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$$



B 都要被指到

onto



$$A \geq B$$

$\geq P \vee Q \vee R$

$\geq P \vee R \vee Q \vee R \vee R$ Resolution $\vdash$

$\wedge \vee$

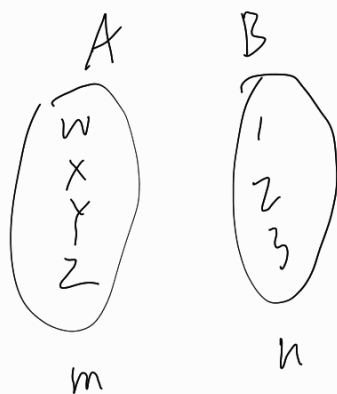
$$\text{all } b \in B \text{ s.t. } f(a) = b$$

map to $|B|^{ |A| } - 2 = 8 - 2 = 6$

map 到 3

$$f: 3 \times 3 \times 3 \times 3 = 3^4 = 81$$

$$\binom{3}{3} 3^4 - \binom{3}{2} 2^4 + \binom{3}{1} 1^4$$



of onto

$$\binom{n}{n} n^m - \binom{n}{n-1} (n-1)^m + \binom{n}{n-2} (n-2)^m = \sum_{k=0}^{n-1} (-1)^k \binom{n}{n-k} (n-k)^m$$

$$= \sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^m$$

$\leftarrow$ onto func. 個數

relation

function

1 to 1

onto

$x \rightarrow 1$
 $y \rightarrow 2$
 $z \rightarrow 3$

$x \rightarrow 1$
 $y \rightarrow 2$
 $z \rightarrow 3$

4

Ex: $A = \{a, b, c, d\}$

$B = \{1, 2, 3\}$

$$3^4 - 3 \cdot 2^4 + 3 \cdot 1^4 = 36$$

4 distinct objects $\rightarrow$ 3 distinguishable container with no empty container

$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$

$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$$\frac{36}{3!} = 6$$

表示

$$\frac{\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m}{n!} = S(m, n)$$

4 distinct obj $\rightarrow$ 3 identical contain with no empty

$$S(4, 4) + S(4, 3) + S(4, 2) + S(4, 1)$$

分4堆

分3堆

分2堆

分1堆

$$S(m+1, n) = S(m, n-1) + n S(m, n)$$

pf: $A = \{a_1, a_2, \dots, a_m, a_{m+1}\}$ 分n堆?

$\begin{matrix} 1 \text{ 堆} \\ \text{---} S(m, n-1) \end{matrix}$

$\begin{matrix} S(m, n) \times n \\ \text{---} a_{m+1}, \end{matrix}$

$$f: A \times A \rightarrow B$$

binary operator

$$f(a, b) = f(b, a)$$

commutative

$$B \subseteq A$$

closed

$$f(a, f(b, c)) = f(f(a, b), c)$$

associative

$$f(a, x) = f(x, a) = a$$

↑
identity

identity

$$f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$$

$$A = \{1, 2, 3, \dots, 7\}$$

$$g(a, b) = \min\{a, b\} \quad \min\{a, \cancel{b}\}$$

$$g(a, b) = \max\{a, b\} \quad \max\{a, \cancel{b}\}$$

f	x	a	b	c	d
x	x	$f(x, a)$ a	b	c	d
a	$f(a, x)$ a	$f(a, a)$	$f(a, b)$	$f(a, c)$	$f(a, d)$
b	b	$f(b, a)$	$f(b, b)$	$f(b, c)$	$f(b, d)$
c	c	$f(c, a)$	$f(c, b)$	$f(c, c)$	$f(c, d)$
d	d	$f(d, a)$	$f(d, b)$	$f(d, c)$	$f(d, d)$

commutative S^1

S^1

$$\left\{ \begin{array}{l} (A \times A, B_1) \\ (A \times A, B_2) \\ (A \times A, B_3) \\ (A \times A, B_4) \\ (A \times A, B_5) \end{array} \right\}$$



$$f: A \rightarrow B \quad |A| = |B|$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\textcircled{1} f: 1\text{-to-1} \iff \textcircled{2} f: \text{onto}$$

$\Rightarrow$ if $f: 1\text{-to-1}$, then $f: \text{onto}$ prove by contradiction

$$f: 1\text{-to-1} \iff f: \text{onto}$$

if $f: 1\text{-to-1}$ not onto

$$|f(A)| < |B| = |A|$$

image codomain if onto $A \cong B$

$$A \subseteq B$$

$$\sum_{k=0}^n (-1)^k \binom{n}{n-k} (n-k)^n$$

$$\leftarrow f: \text{onto} \rightarrow f: 1\text{-to-1}$$

if f onto but not 1-to-1

$$\exists a_1, a_2 \in A$$

$$f(a_1) = f(a_2)$$

$$|f(A)| < |A| = |B|$$

image codomain

Complexity

program $O(n^2)$

Def $f, g: \mathbb{Z}^+ \rightarrow \mathbb{R}$ $f(n)=n$

$g(n)=n^2$

g dominated f : (f is dominated by g)

$\exists m \in \mathbb{R}^+ \quad k \in \mathbb{Z}^+$

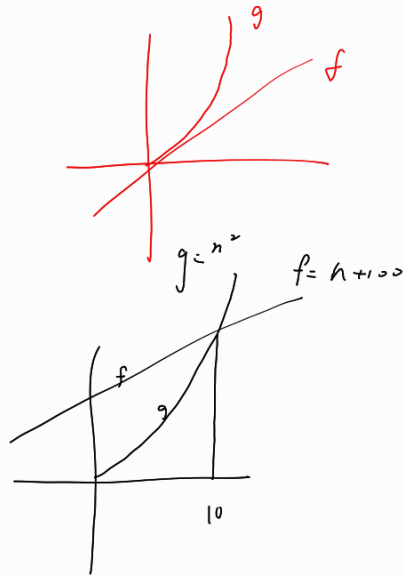
s.t. $|f(n)| \leq m |g(n)|$

$n \in \mathbb{Z}^+ \quad n \geq k$

f is dominated by g

f is order g

$f \in O(g)$



$O(1)$

$O(\log n)$

$O(n)$

$O(n \log n)$

$O(n^2)$